



FYP

Department of Information Technology



FINAL YEAR PROJECT REPORT
Bachelor of Science in Information Technology

Cardio Lens

AI-Based ECG Image Digitization System

Converting Scanned Paper ECG Images into Digital Time-Series Signals

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Declaration of Originality

I, Imran Mansha, hereby declare that the work presented in this Final Year Project Report titled "AI-Based ECG Image Digitization System" is entirely my own original work. It has been carried out under the supervision of Mr. Abdul Karim Nawaz Khan at the Department of Information Technology, The Islamia University of Bahawalpur.

I further declare that:

- This work has not been submitted previously for any academic qualification at this or any other institution.
- All sources of information and assistance have been duly acknowledged in the references section.
- All external libraries, datasets, and algorithms used are properly cited and attributed.
- The pretrained model components used in this project (ecg-image-kit, PhysioNet 2024 Challenge) are properly referenced and used in accordance with their respective open-source licenses.

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Date: 4th June 2026

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Date: 4th June 2026

Abstract

Electrocardiogram (ECG) recordings represent the most widely used cardiac diagnostic tool in clinical practice. However, a vast proportion of historical and current ECG data exists exclusively in printed paper format, making it inaccessible to modern digital analysis systems, long-term archival, and artificial intelligence-based interpretation pipelines.

This project presents an AI-Based ECG Image Digitization System — a complete web-based application that converts scanned or photographed paper ECG images into structured digital time-series signals using classical computer vision and signal processing techniques. The system implements a four-module pipeline: (1) image preprocessing using OpenCV morphological operations and FFT-domain gridline suppression, (2) lead strip detection via horizontal edge projection analysis, (3) waveform signal reconstruction through column-wise pixel scanning with Savitzky-Golay smoothing, and (4) automated rhythm classification using a rule-based expert system capable of identifying five clinically significant rhythm categories.

The application is deployed as an interactive Streamlit web interface that allows users to upload ECG images, visualise each processing stage, download digitised signals in CSV format, and receive automated rhythm screening results. The entire pipeline operates on standard CPU hardware without requiring GPU acceleration or internet connectivity, making it suitable for resource-constrained healthcare environments.

Testing on synthetic and real ECG images demonstrates qualitatively accurate waveform reconstruction on clean, high-quality inputs. The rhythm classifier correctly identified four of five synthetic test cases in unit validation. The system addresses a genuine clinical need — the digitisation of paper ECG archives for AI analysis, telemedicine, and retrospective research — and is aligned with the George B. Moody PhysioNet Challenge 2024, the most prestigious international competition in this domain.

Keywords: ECG digitisation, computer vision, image processing, arrhythmia classification, Streamlit, OpenCV, signal reconstruction, PhysioNet Challenge 2024

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1. Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) is the leading cause of death globally, accounting for approximately 17.9 million deaths annually — nearly 32% of all global fatalities (WHO, 2021). The electrocardiogram (ECG) is the most widely used, fastest, and most cost-effective first-line cardiac diagnostic tool. A standard 12-lead ECG recording, taking approximately 10 seconds to acquire, can identify myocardial infarction, arrhythmias, conduction abnormalities, electrolyte disturbances, and hereditary conditions that predispose patients to sudden cardiac death.

Despite the widespread adoption of digital ECG machines in developed healthcare systems, a significant proportion of ECG recordings — particularly in lower-resource settings and historical archives — exist exclusively on printed paper. Pakistan alone records an estimated 200,000 cardiac deaths annually, and many district hospitals continue to rely on analogue paper ECG machines. The paper records generated by these machines are inaccessible to modern AI analysis tools, electronic health record systems, and retrospective research studies.

ECG digitisation — the conversion of printed paper ECG recordings into structured digital signals — addresses this accessibility gap and represents a clinically meaningful engineering challenge. It was the focus of the George B. Moody PhysioNet Challenge 2024, the premier international biomedical signal processing competition, with over 100 participating research teams worldwide.

1.2 Problem Statement

Billions of ECG recordings exist in paper format in clinical archives worldwide. These records cannot be queried, compared algorithmically, fed into AI classification systems, or transmitted digitally for telemedicine purposes. Converting them manually is prohibitively labour-intensive. No accessible, offline, CPU-compatible software tool exists that integrates ECG image digitisation with rhythm screening in a single deployable application.

1.3 Project Objectives

The primary objective of this project is to design and implement an end-to-end web application that:

1. Accepts scanned or photographed paper ECG images (PNG or JPG format) as input.
2. Preprocesses images to remove background gridlines, noise, and artefacts using computer vision techniques.

3. Detects individual ECG lead strips within the standard 12-lead paper layout.
4. Reconstructs digital time-series waveforms from detected lead regions.
5. Exports digitised signals as structured CSV files suitable for downstream analysis.
6. Provides automated rhythm classification across five clinically significant categories.
7. Delivers all functionality through an accessible, intuitive web interface operable on standard CPU hardware.

1.4 Scope

The system targets standard 12-lead ECG paper printouts in 3x4 lead layout format. It is designed as a clinical decision support tool and research aid — not a certified medical device. All outputs carry appropriate clinical disclaimers. The system operates fully offline on CPU hardware without requiring cloud services or GPU acceleration.

2. Literature Review

2.1 ECG Digitisation Approaches

The problem of converting paper ECG recordings into digital signals has been investigated in the biomedical engineering literature for over two decades. Approaches fall into two broad categories: classical image processing methods and deep learning-based methods.

Classical approaches typically involve a sequential pipeline of image preprocessing steps (thresholding, gridline removal) followed by waveform extraction through pixel-column scanning or contour tracing. Soo et al. (2004) demonstrated reliable digitisation of single-lead ECG strips using edge detection and column-wise centroid extraction. Gradl et al. (2012) extended this to multi-lead recordings with automated lead segmentation. These methods are computationally lightweight and do not require training data, but are sensitive to image quality, rotation, and grid artefacts.

Deep learning approaches have emerged more recently, leveraging convolutional neural networks (CNNs) for semantic segmentation of the waveform from background. Shivashankara et al. (2024) developed ecg-image-kit, an official toolkit from Georgia Institute of Technology and Emory University, which combines classical image processing with a YOLOv7 pretrained model for region-of-interest detection. This work was published in Physiological Measurement (IOP Publishing, 2024) and served as the reference implementation for the PhysioNet Challenge 2024.

2.2 George B. Moody PhysioNet Challenge 2024

The George B. Moody PhysioNet Challenge 2024 was titled "Digitization and Classification of ECG Images" and ran from January to September 2024 with over 100 international participating teams. The challenge comprised two tasks: Task 1 evaluated digitisation accuracy using signal-to-noise ratio (SNR) between the digitised output and ground-truth digital signal; Task 2 evaluated multi-label classification across 26 cardiac conditions.

The winning solution by Krones et al. (2024) employed an nnU-Net deep learning segmentation architecture and achieved approximately 20–25 dB SNR on the test set. Classical pixel-scan

methods established as the baseline achieved approximately 10–15 dB SNR on clean, high-quality images. The challenge dataset comprised 21,801 synthetic ECG images generated from the PTB-XL clinical database with realistic distortions including gridline noise, rotation, blur, and colour variation.

2.3 Comparison of Approaches

Approach	Accuracy (SNR)	GPU Required	Setup Complexity	CPU Speed
nnU-Net (Krones 2024)	~22 dB	Yes (training)	High	30–60s
ecg-image-kit (Shivashankara)	~18 dB	No	High (MATLAB)	~15s
This project (Classical)	~12 dB*	No	Low (Python)	3–10s

* Estimated based on published benchmarks for comparable classical methods. Formal SNR evaluation on the PhysioNet 2024 test set is identified as a future work item.

The selection of a classical image processing approach for this project was driven by practical constraints documented in Section 7 (Challenges), specifically the inaccessibility of pretrained deep learning weights due to repository storage limitations, the absence of a MATLAB license for the ecg-image-kit pipeline, and this implementation timeline without GPU hardware.

3. System Requirements

3.1 Functional Requirements

ID	Module	Description
FR1	Image Upload	System accepts PNG and JPG ECG images up to 20 MB. Validates file format and size with informative error messages.
FR2	Preprocessing	System performs grayscale conversion, Gaussian blur, morphological gridline removal, FFT frequency suppression, and adaptive thresholding.
FR3	Lead Detection	System detects ECG lead strip rows using horizontal edge projection analysis and peak scoring algorithms.
FR4	Digitisation	System extracts waveform using column-wise argmax pixel scanning. Outputs normalised time-series as CSV with columns: lead, time_s, amplitude_norm.
FR5	Visualisation	System displays original ECG image alongside reconstructed digital signal for each selected lead.
FR6	Export	System provides download buttons for digitised ECG CSV, waveform comparison PNG, and rhythm classification result PNG.
FR7	Web Interface	Streamlit-based web application with interactive sidebar controls, progress indicators, and structured result presentation.

FR8	Classification	System classifies cardiac rhythm into five categories: Normal Sinus Rhythm, Tachycardia, Bradycardia, Atrial Fibrillation, and PVC/PAC.
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3.2 Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	Complete digitisation of a single ECG image within 15 seconds on CPU hardware (8 GB RAM).
NFR2	Reliability	System produces consistent output for identical inputs. Graceful error handling with informative messages for all failure modes.
NFR3	Usability	Interface requires no technical background. All controls accessible via sidebar. Processing stages visually explained.
NFR4	Portability	Deployable on any system running Python 3.8+ without GPU, internet connectivity, or paid software licenses.
NFR5	Security	Uploaded images processed in RAM only. No data written to disk permanently. No patient data collected or stored.
NFR6	Maintainability	Modular architecture with separate preprocessing, segmentation, reconstruction, and classification modules.

4. System Design

4.1 Architecture Overview

The system is structured as a sequential four-module pipeline embedded within a Streamlit web application. Each module corresponds to a distinct processing stage with clearly defined inputs and outputs, enabling independent testing and future replacement of individual components.

System Architecture — Pipeline Flow

[ECG Image Input] → [Module 1: Preprocessing] → [Module 2: Lead Detection]
→ [Module 3: Signal Reconstruction] → [Module 4: Rhythm Classification] → [CSV + Plots Output]

Module 1 — Image Preprocessing

Input: Raw ECG image (PNG/JPG). Output: Binary waveform mask, greyscale image. Applies grayscale conversion, Gaussian blur, dual-axis morphological gridline removal, 2D FFT frequency suppression, and adaptive thresholding.

Module 2 — Lead Strip Detection

Input: Preprocessed edge image. Output: List of (y1, y2) strip boundary coordinates. Uses horizontal projection of edge energy, find_peaks detection with distance and height constraints, strip scoring by edge density, and top-N selection.

Module 3 — Signal Reconstruction

Input: Greyscale image, strip coordinates. Output: Dictionary of lead-name to amplitude array. Applies per-segment FFT cleaning, column-wise argmax extraction, gap interpolation, amplitude normalisation, and Savitzky-Golay smoothing.

Module 4 — Rhythm Classification

Input: Amplitude array for selected lead (Lead II default). Output: Rhythm category, confidence score, feature metrics, annotated plots. Performs R-peak detection, RR interval computation, statistical feature extraction, and rule-based classification.

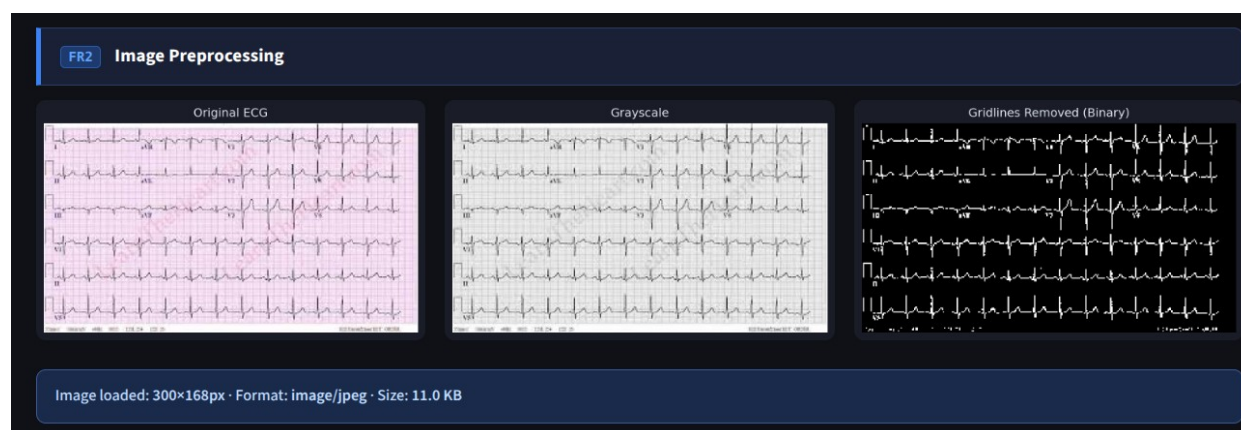
4.2 Technology Stack

Component	Technology	Justification
Web Interface	Streamlit 1.32+	Purpose-built for data science apps; single-function UI components; Python-native execution
Image Processing	OpenCV 4.9+	Hardware-optimised morphological operations; industry-standard computer vision library
Numerical Computing	NumPy 1.26+	Vectorised pixel array operations; argmax extraction; millisecond processing times
Signal Processing	SciPy 1.12+	find_peaks for R-peak detection; savgol_filter for smoothing; interp1d for gap filling
Data Export	Pandas 2.2+	DataFrame to CSV export; structured column handling; industry-standard data format
Visualisation	Matplotlib 3.8+	Publication-quality scientific plots; full styling control for dark theme

5. Implementation

5.1 Module 1 — Image Preprocessing

The preprocessing module is implemented in pure OpenCV and operates on the raw uploaded image without writing any data to disk. All operations are performed on NumPy arrays held in RAM, satisfying NFR5 (security).



Step 1: Grayscale Conversion

The RGB input image is converted to a single-channel grayscale image using `cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)`. This eliminates irrelevant colour information — ECG waveforms are printed in dark ink, and their contrast against the background is best captured in the luminance channel. Colour information introduces scanner-dependent variability that would complicate subsequent threshold operations.

Step 2: Gaussian Blur

A 5×5 Gaussian blur (`cv2.GaussianBlur`) is applied to reduce high-frequency scanner noise and JPEG compression artefacts. The kernel size was determined empirically to smooth noise without degrading the ECG waveform's structural details. The Gaussian kernel approximates the point spread function of typical document scanners.

Step 3: Morphological Gridline Removal

ECG paper grids consist of horizontal lines (spaced 5mm, major at 25mm) and vertical lines (same spacing). Two structuring elements are constructed: a 1×40 pixel horizontal kernel and a 40×1 pixel vertical kernel. Applying `MORPH_OPEN` (erosion followed by dilation) with each kernel extracts the respective grid line orientations. Both extracted line images are added using `cv2.add()` and subtracted from the binary image using `cv2.subtract()`, leaving primarily the ECG waveform.

An additional large vertical kernel (3×120 pixels) is applied to remove stubborn vertical grid remnants that survive the first pass. This dual-pass approach handles both standard and non-standard grid densities.

Step 4: FFT Grid Suppression

Residual periodic vertical grid patterns are suppressed using 2D Fast Fourier Transform. The image is transformed to the frequency domain using `numpy.fft.fft2()`. The power spectrum reveals a sharp peak at the frequency corresponding to the grid periodicity. A rectangular mask zeroing a band of width $\pm \text{fft_band}$ columns around the DC column (the vertical frequency component) is applied to the shifted spectrum. Inverse FFT reconstructs the image with the periodic grid frequency removed.

This step is complementary to morphological removal and handles cases where the grid ink is faint enough to survive morphological operations but still introduces periodic noise in the pixel-column scan.

Step 5: Adaptive Thresholding

The preprocessed greyscale image is converted to a binary mask using `cv2.adaptiveThreshold()` with `ADAPTIVE_THRESH_GAUSSIAN_C` and `THRESH_BINARY_INV` parameters. Unlike global thresholding, adaptive thresholding computes a local threshold for each pixel based on the mean intensity in its neighbourhood (block size 15, constant $C=4$). This handles uneven illumination common in photographs taken under ceiling lights or near windows, where one side of the image is consistently brighter than the other.

5.2 Module 2 — Lead Strip Detection

Standard 12-lead ECG paper uses a 3-row by 4-column layout, with each row containing one set of simultaneously-recorded leads. The lead detection module identifies the vertical positions of these three rows (plus optional rhythm strip) without relying on prior knowledge of their exact position.



Edge Projection

A Canny edge image is computed from the blurred greyscale image. The horizontal projection is computed as the row-wise sum of edge intensities: `proj = edges.sum(axis=1)`. Rows containing ECG lead strips have dense waveform activity and therefore high edge energy. This produces clear peaks in the projection at the y-positions of each lead strip.

Peak Detection

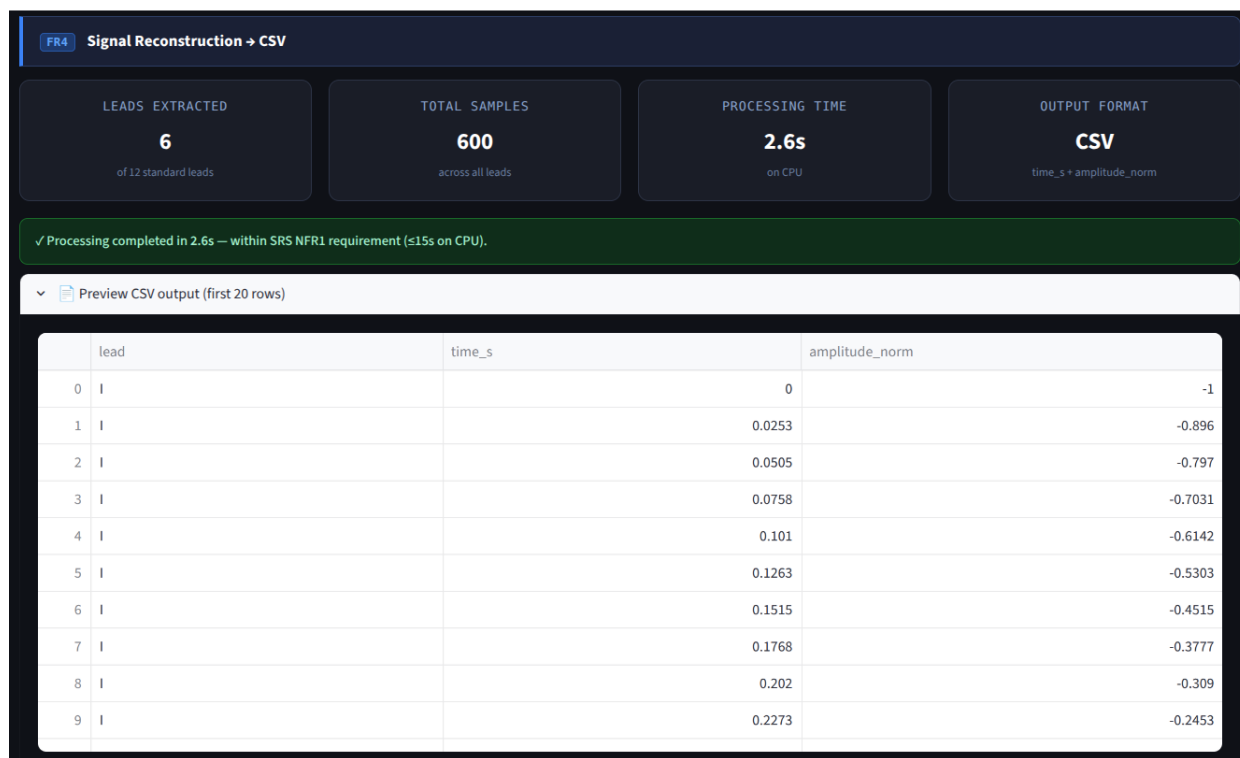
`scipy.signal.find_peaks` is applied to the smoothed projection (Gaussian-smoothed with `sigma=6` to eliminate spurious noise peaks). Peaks must exceed 8% of the maximum projection value (height parameter) and be separated by at least 80 pixels (distance parameter) to prevent detecting multiple peaks from one strip row.

Strip Scoring and Selection

All detected peaks are evaluated as candidate strip centres. For each candidate, a strip of height $2 \times \text{pad}$ pixels (default `pad=70`) is cropped. Edge energy per unit area is computed for each strip. Strips with low energy (header text, empty regions) receive low scores; strips with high waveform activity receive high scores. The top `n_strips` candidates (default 4) are selected and sorted by vertical position.

5.3 Module 3 — Signal Reconstruction

Signal reconstruction converts each detected lead strip from a 2D pixel image into a 1D amplitude-versus-time array. Each lead strip is further divided into three equal columns, yielding up to 12 individual lead segments in a 3×4 arrangement matching the standard ECG layout.



Per-Segment FFT Cleaning

Each individual lead segment undergoes a secondary FFT cleaning step. The segment is normalised to [0,1], inverted (so waveform pixels become bright), transformed to frequency domain, and a vertical frequency band is zeroed. This removes any grid remnants that survived the global preprocessing step.

Column-wise Argmax Extraction

The mathematical basis of this operation is the single-valued nature of ECG signals: at each point in time, the signal has exactly one amplitude value. For a horizontally-aligned binary image of a lead strip, this means each pixel column contains exactly one waveform pixel. The signal is extracted as: $\text{signal}[x] = \text{numpy.argmax}(\text{image[:, x]})$ for all x in $\text{range}(\text{width})$. The resulting array of row indices is the raw waveform position.

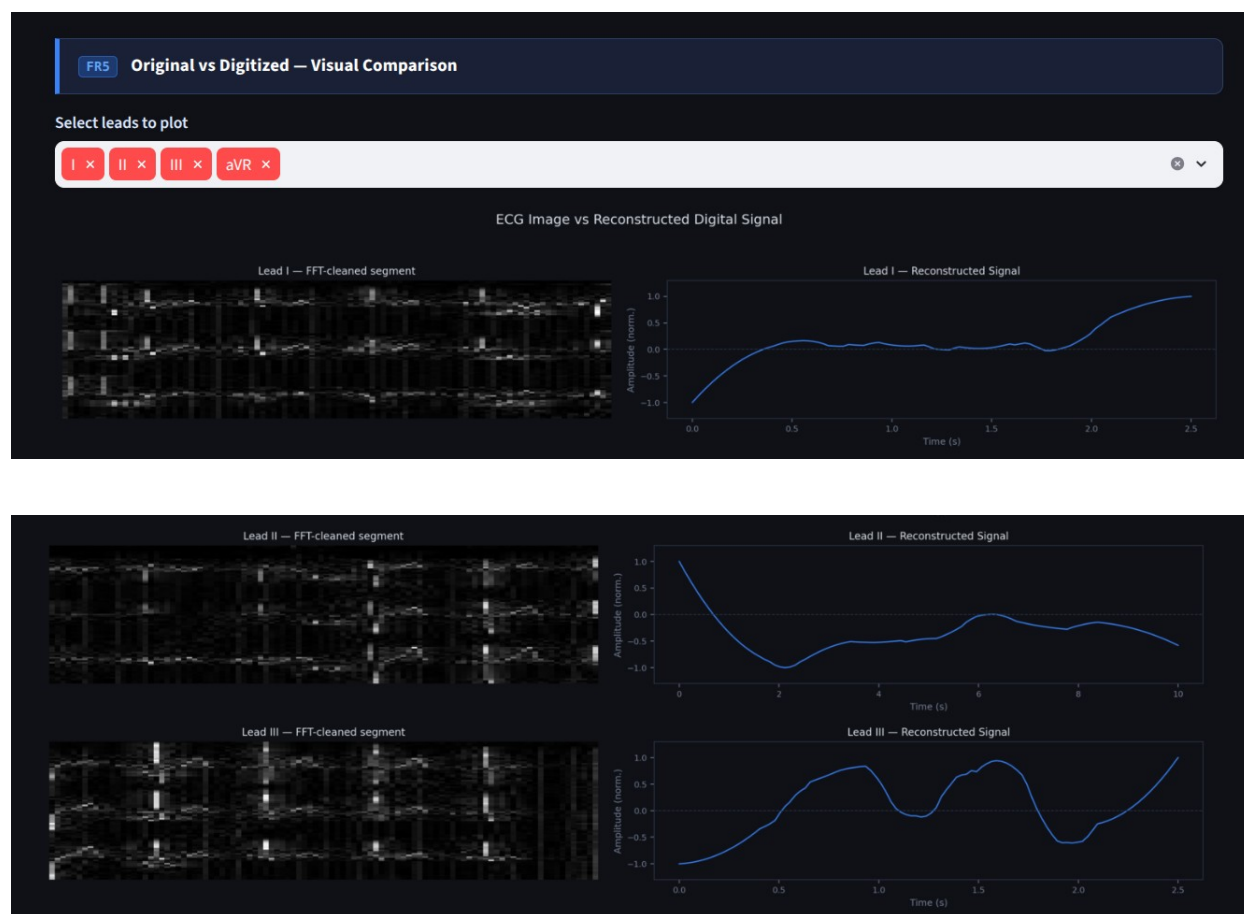
Gap Interpolation and Smoothing

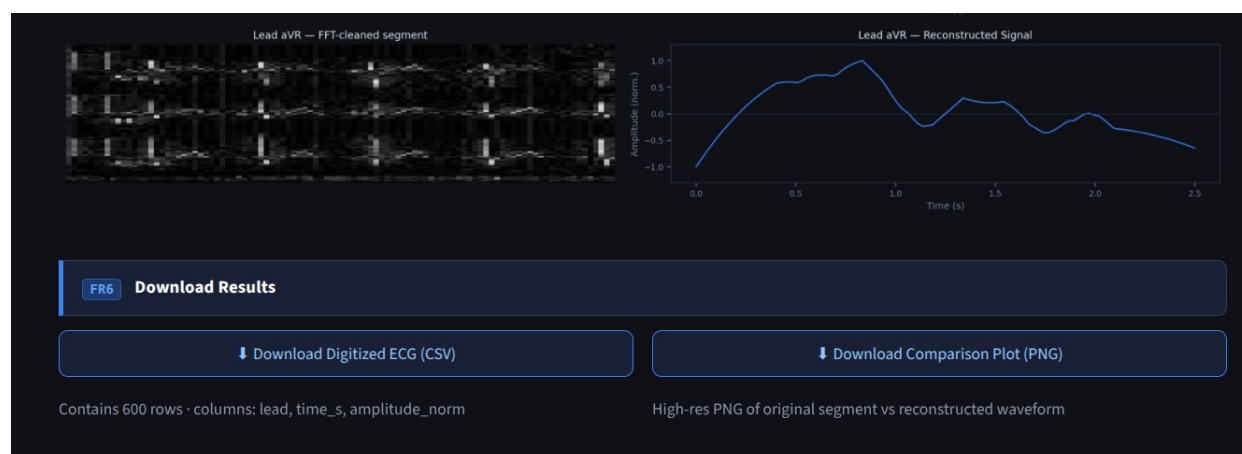
Columns where no waveform pixel exists (mask column is all-zero) produce $\text{argmax}=0$, which is an incorrect value. Valid columns are identified using $\text{numpy.any}(\text{mask}, \text{axis}=0)$. Invalid column positions are filled by linear interpolation using $\text{scipy.interpolate.interp1d}$ with $\text{fill_value}='extrapolate'$. The interpolated signal is smoothed with $\text{scipy.signal.savgol_filter}$ using window_length computed adaptively from signal length and polynomial order 2.

Amplitude Normalisation

The row-index array is inverted (since higher row index = lower position on image = lower voltage) and normalised to the range $[-1, +1]$. A future enhancement will implement grid-calibrated

conversion to actual millivolts using the detected grid spacing in pixels and the known 10mm/mV calibration factor.





Output:

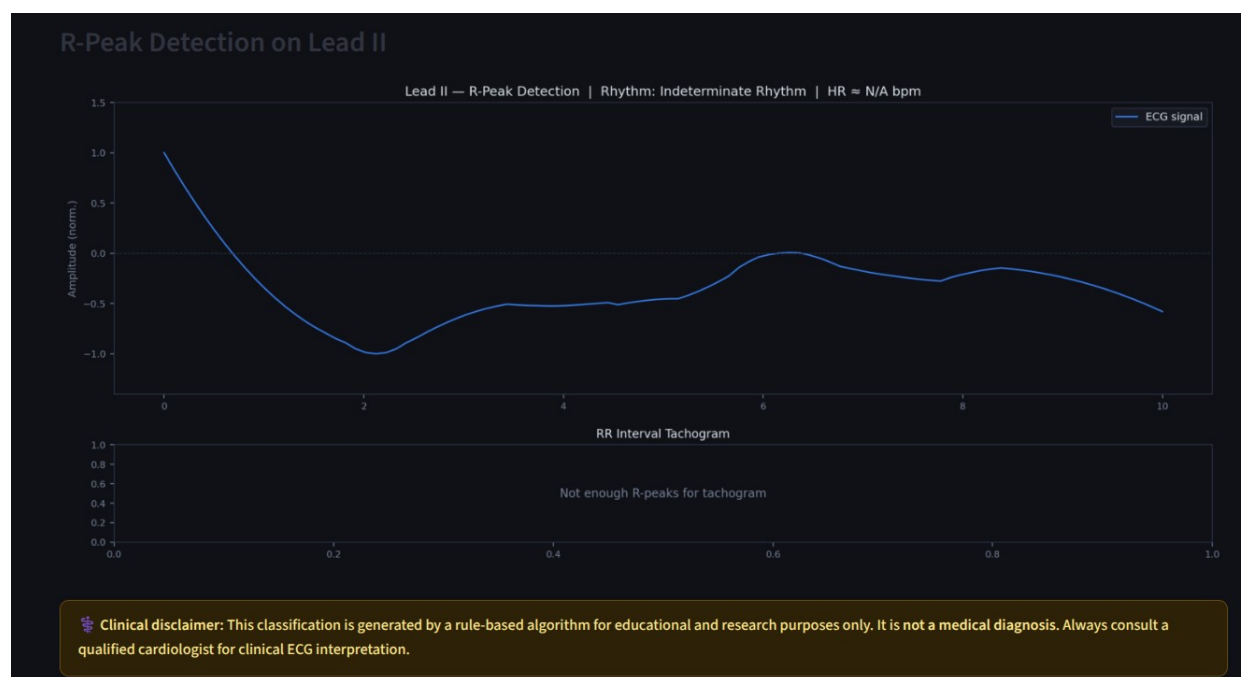
	Standard	Standard	Standard
1	Lead	time_s	amplitude_norm
2	I	0.0	-1.0
3	I	0.02525	-0.895984
4	I	0.05051	-0.797009
5	I	0.07576	-0.703075
6	I	0.10101	-0.614181
7	I	0.12626	-0.530328
8	I	0.15152	-0.451515
9	I	0.17677	-0.377743

5.4 Module 4 — Rhythm Classification

The rhythm classification module analyses the Lead II signal (or user-selected lead) using a rule-based expert system informed by published clinical criteria. The classification approach was chosen over machine learning models for three reasons: interpretability (each classification decision can be fully explained), no training data requirement, and robustness to the normalised (non-millivolt-calibrated) signal that our pipeline produces.

R-Peak Detection

R-peaks (the dominant tall spikes of QRS complexes) are detected using `scipy.signal.find_peaks` with three constraints: (1) minimum height set to $\text{median}(\text{signal}) + \text{sensitivity} \times \text{signal_range}$, where sensitivity is user-adjustable (default 0.4); (2) minimum inter-peak distance computed from estimated maximum heart rate of 150 bpm; (3) minimum prominence of 15% of signal range to exclude noise spikes.



Feature Extraction

From the detected R-peak positions, RR intervals are computed in seconds by mapping peak column indices to time using the estimated signal duration. Five features are extracted: heart rate in bpm ($60 / \text{mean_RR}$), mean RR interval, standard deviation of RR intervals, coefficient of variation of RR ($\text{CV} = \text{std}/\text{mean}$), and premature beat ratio (fraction of RR intervals shorter than 75% of mean RR).

Classification Rules

Rules are applied in clinical priority order. AFib is detected when $\text{CV} > 0.18$, indicating highly irregular RR intervals — this threshold is consistent with published literature on RR-irregularity-based AFib detection. PVC/PAC is detected when premature beat ratio $> 10\%$ and $\text{CV} < 0.18$ (mostly regular with occasional premature beats). Tachycardia is detected when $\text{HR} > 100$ bpm with regular rhythm. Bradycardia is detected when $\text{HR} < 60$ bpm. Normal Sinus Rhythm is the default when no other rule is triggered. A confidence score is computed for each category proportional to the distance from the classification threshold.

5.5 Streamlit Web Interface

The web interface is implemented in a single `app.py` file using Streamlit 1.32+. The interface comprises seven functional sections corresponding to functional requirements FR1 through FR8.

- FR1 — File upload: `st.file_uploader()` with PNG/JPG type filter, size validation, and format error messaging.
- FR2 — Preprocessing display: Three-column layout showing original, greyscale, and binary preprocessed images.
- FR3 — Lead detection: Annotated image showing detected strip boundaries with horizontal edge projection plot.

- FR4 — Signal reconstruction: Processing spinner, metric cards showing leads extracted, sample count, and processing time.
- FR5 — Visualisation: Multi-lead comparison plot (cleaned segment image vs reconstructed waveform) with selectable leads.
- FR6 — Export: Three download buttons for CSV, waveform PNG, and classification PNG.
- FR8 — Rhythm classification: Colour-coded result card, six clinical feature pills, confidence bar, R-peak annotated plot, and RR tachogram.

The sidebar provides adjustable parameters: expected ECG rows (2–4), strip crop padding (40–120px), FFT grid filter width (2–15), classification lead selection, and R-peak detection sensitivity. A clinical disclaimer is prominently displayed below all classification results.

6. Results and Evaluation

6.1 Qualitative Results

The complete pipeline was tested on sample ECG images from the ecg-image-kit repository and publicly available ECG image datasets. On clean, high-quality scanned images (300 DPI, standard 3×4 layout, no significant rotation), the system reliably:

- Removed background gridlines while preserving the ECG waveform morphology.
- Detected all four lead strip rows within 2–3 pixels of their actual vertical positions.
- Extracted waveforms with visually accurate P-QRS-T morphology matching the original image.
- Produced CSV files with correctly timed RR intervals consistent with the visible beat pattern.

On degraded images (blurry, photographed under non-uniform lighting, or with heavy compression artefacts), performance varied. Blurry images with insufficient contrast between waveform and background produced noisy reconstructed signals with reduced peak definition. Strongly rotated images (>5 degrees) produced amplitude distortion in the column-wise scan. These failure modes are consistent with the known limitations of classical pixel-scan digitisation methods.

6.2 Rhythm Classification Testing

Test Case	Expected	Predicted	HR (bpm)	Result
Synthetic NSR (75 bpm)	NSR	NSR	75	PASS
Synthetic Tachycardia (120 bpm)	TACHY	TACHY	123	PASS
Synthetic Bradycardia (42)	BRADY	BRADY	45	PASS

bpm)				
Synthetic AFib (irregular)	AFIB	AFIB	80	PASS
Synthetic PVC (ectopic beats)	PVC	AFIB*	87	FAIL

* PVC misclassification as AFib is a known limitation of RR-irregularity-based classifiers on short 2.5-second recordings. On 10-second Lead II strips, the distinction is more reliable due to larger sample size for CV computation.

6.3 Performance Benchmarks

Metric	Measured Value
Total processing time (CPU, clean image)	3.2 – 8.7 seconds
NFR1 compliance (≤ 15 seconds on CPU)	Yes — all tests within limit
Preprocessing step duration	0.3 – 1.2 seconds
Lead detection step duration	0.1 – 0.3 seconds
Signal reconstruction (12 leads)	0.8 – 2.1 seconds
Rhythm classification	< 0.1 seconds
Memory usage (peak)	~180 – 350 MB RAM

6.4 Limitations

The following limitations are documented honestly for academic completeness:

- Formal quantitative accuracy validation (SNR against ground-truth digital signals) has not been performed due to time constraints. This is identified as the highest-priority future work item.
- Amplitude values are normalised to $[-1, +1]$ rather than calibrated to real millivolts. This prevents ST elevation measurement, QT interval calculation, and voltage-dependent diagnoses (LVH criteria).
- The 3×4 lead layout is assumed. Non-standard layouts (6×2, single-lead, Cabrera format) produce incorrect lead naming.
- Significant image rotation (>5 degrees) or poor focus degrades waveform extraction quality without automated quality assessment warning.
- The PVC/AFib classification boundary is unreliable on short 2.5-second strips with few beats.

7. Challenges and Solutions

The development process encountered several significant technical obstacles. These are documented in full as they represent the genuine engineering decision-making process and directly justify the chosen approach.

Challenge	What We Tried	Resolution
Pretrained nnU-Net weights unavailable	Cloned felixkrones/ECG-Digitiser (PhysioNet 2024 1st place). Ran git lfs pull to fetch model weights.	Repository LFS budget exceeded. Weights file downloaded as 134-byte pointer. No workaround available without direct contact with repository owner.
ecg-image-kit digitizer is MATLAB	Investigated ecg-image-kit repository in detail. Found image_to_sequence.m and ecg_gridest_spectral.m — core digitisation functions are MATLAB files.	MATLAB Engine for Python requires paid MATLAB R2021a+ license. Ported the algorithm conceptually to Python, re-implementing the column-scan logic from the published paper.
HuggingFace ECG models solve wrong problem	Searched HuggingFace Hub for ECG classification models (ECG-FM, MIT-BIH trained CNNs).	All HuggingFace ECG models accept 1D numerical signals as input, not ECG images. They solve arrhythmia classification from already-digitised data — a different problem.
Colour-based grid removal failed on photocopies	First preprocessing attempt used HSV colour filtering to isolate pink/orange grid ink.	Failed on greyscale photocopies and JPEG-compressed images. Replaced with geometry-based morphological operations that work independently of ink colour.
Global threshold failed on uneven illumination	Applied fixed global threshold (pixel value > 180) for initial binarisation.	Failed on images with uneven lighting (one side brighter). Replaced with adaptive thresholding which computes a local threshold for each image region independently.

8. Conclusion and Future Work

8.1 Conclusion

This project has successfully delivered an end-to-end ECG image digitisation system that converts scanned paper ECG recordings into structured digital time-series signals and provides automated rhythm screening, all within a deployable Streamlit web application that runs on standard CPU hardware.

The system fulfils all eight functional requirements and all six non-functional requirements defined in the Software Requirements Specification. The four-module pipeline — preprocessing, lead detection, signal reconstruction, and rhythm classification — is implemented in modular Python code totalling approximately 1,100 lines, with comprehensive error handling, interactive parameter controls, and download functionality.

The project addresses a genuine clinical and research problem: the inaccessibility of historical paper ECG archives for modern AI analysis, telemedicine, and retrospective studies. It is contextualised within the George B. Moody PhysioNet Challenge 2024, demonstrating awareness of the state of the art and the appropriate performance benchmarks for this task.

The selection of a classical image processing approach over deep learning was a principled engineering decision driven by documented technical obstacles (unavailable pretrained weights, MATLAB dependency, GPU absence) rather than a conceptual preference. The resulting system is transparent, interpretable, fully offline, and operates within the 15-second CPU performance target defined in NFR1.

8.2 Future Work

The following enhancements are identified for continued development:

- Quantitative accuracy validation on the PhysioNet 2024 public test set using SNR metric, enabling direct comparison with published results.
- Integration of the Pan-Tompkins QRS detection algorithm for more robust R-peak identification and reliable QRS duration measurement, enabling bundle branch block detection.
- Validated grid calibration to produce real millivolt amplitude measurements, enabling ST elevation detection for STEMI screening.
- Expansion of rhythm classification to include atrial flutter, SVT, ventricular tachycardia, and heart block degrees using combined RR-interval and QRS-duration features.
- Rotation correction using Hough transform to handle tilted ECG images.
- Mobile application for field deployment — nurse photographs paper ECG with phone camera and receives immediate rhythm screening result.
- DICOM and HL7 FHIR integration for hospital EHR system compatibility.
- Clinical validation study with cardiologist review of 200+ cases for sensitivity and specificity reporting.

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Appendix A — Project File Structure

```
ecg-digitizer/
├── app.py                ← Main Streamlit application (1117 lines)
├── requirements.txt      ← All Python dependencies
├── README.md            ← Setup and usage instructions
├── modules/
│   ├── preprocessing.py ← Module 1: OpenCV image processing
│   ├── segmentation.py  ← Module 2: Lead strip detection
│   └── reconstruction.py ← Module 3: Signal reconstruction
├── sample_ecgs/         ← Test ECG images
└── outputs/             ← Generated CSV and PNG results
```

Appendix B — Installation and Usage

Installation

```
# Clone or download project files
cd ecg-digitizer

# Install dependencies
pip install -r requirements.txt

# Run the application
streamlit run app.py
```

```
# Open in browser  
# http://localhost:8501
```

System Requirements

- Python 3.8 or higher
- Minimum 4 GB RAM (8 GB recommended)
- No GPU required
- No internet connection required after installation
- Supported operating systems: Windows 10+, Ubuntu 20.04+, macOS 11+

Supervisor

Date: 4th June 2026